Chapter 03

Networking & The Linux Shell

Episode 3.01

Episode title: **Network Basics**

Objective: 1.4

Linux Network Fundamentals

Network interfaces

IP addressing

Network configuration layers

Network Interfaces in Linux

- ip addr
- ip link
- nmcli

NetworkManager Tools (Exam)

NetworkManager manages network connections

`nmcli` Command-line management tool

`nmconnect` Establish and manage connections

Used on desktop and server systems

Netplan Overview (Ubuntu)

Netplan is used on Ubuntu systems

YAML-based network configuration

Configuration files in `/etc/netplan`

Works with NetworkManager or `systemd-networkd`

Netplan Commands (Exam)

`netplan apply` – Apply configuration

`netplan status` – View current state

`netplan try` – Test configuration safely

Used after editing Netplan files

Hostname Resolution

/etc/hosts

DNS

systemd-resolved

Name Resolution Configuration Files

`/etc/hosts` – Local hostname mappings

`/etc/resolv.conf` – DNS resolver configuration

`/etc/nsswitch.conf` – Name resolution order

Exam focus: which file is checked first

Basic Network Configuration (DEMO)

View interfaces

Configure IP

Verify connectivity

Episode 3.02

Episode title: **Network Utilities**

Objective: 1.4

A Structured Approach to Network Troubleshooting

Connectivity checks

Path analysis

Local services and ports

Introduction to Packet Inspection

tcpdump

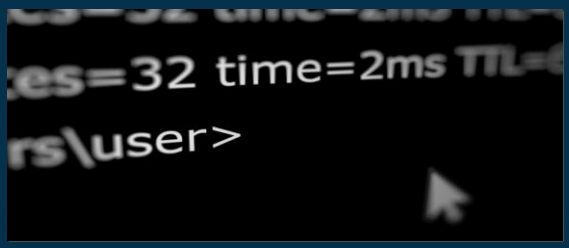
Observing real traffic

Network Troubleshooting Utilities (DEMO)

- ping
- traceroute
- ss/netstat
- tcpdump

Core Network Utilities

- ping
- traceroute
- ss/netstat



Common Network Tools – Exam Coverage

Addressing & interfaces:

arp, ethtool, hostname, ip

Connectivity & testing:

ping, ping6, mtr, tracepath, traceroute

Name resolution & services:

dig, nslookup, curl, nc

Episode 3.03

Episode title: **Advanced Network Utilities**

Objective: 1.4

Deep Packet Analysis Tools

tcpdump

Wireshark

tshark

Network Scanning and Discovery

nmap

Port scanning

Service detection

Network Performance Testing

iperf3

Throughput measurement

Latency and bandwidth

Advanced Network Troubleshooting (DEMO)

tcpdump

nmap

iperf3

Episode 3.04

Episode title: **Shell Environment**

Objective: 1.5

Shell Types and Initialization

Shell types

Login vs non-login shells

Interactive vs non-interactive

Shell Configuration Files

~/.bashrc

~/.bash_profile

/etc/profile

Absolute vs Relative Paths

Absolute paths:

Begin at `root/`

`~` represents home directory

Relative paths:

`.` current directory

`..` parent directory

`-` previous directory

Shell Configuration Files – When They Apply

`.bashrc` – Interactive non-login shells

`.bash_profile` – Login shells

`.profile` – System-wide or fallback

Determines when variables and aliases load

Environment Variables

Defining variables

Exporting variables

Usage in commands

Common Environment Variables

- HOME – User home directory
- PATH – Command search path
- USER – Current username
- SHELL – Default shell
- DISPLAY – Graphical display
- PS1 – Shell prompt format

Episode 3.05

Episode title: **Shell Environment Demo**

Objective: 1.5

Shell Environment Configuraion (DEMO)

Set variables

Modify profiles

Test behavior

Episode 3.06

Episode title: **Essential Shell Utilities**

Objective: 1.5

Text Processing Fundamentals

Text streams

Standard input/output

Pipelines

Shell I/O Streams

Standard input stdin

Standard output stdout

Standard error stderr

Streams can be redirected

Channel Redirections Operators

Input: `<`

Output: `>, >>`

Pipes: `|`

Here documents: `<< , <<<`

Redirecting `stderr`

Core Text Processing Utilities

- grep
- sed
- awk
- sort
- cut

Basic Shell Utilities – Exam Set

File & output:

cat, echo, printf, tee

Viewing:

head, tail, less, more

Text processing:

tr, uniq, wc, xargs

Shell Productivity Commands

Command history:

history, !, !!

Customization:

alias, source

System information:

uname

Calculation:

bc

Real-World Use Cases

Log analysis

Configuration review

Automation

Episode 3.07

Episode title: **Essential Shell Utilities Demo**

Objective: 1.5

Essential Shell Utilities (DEMO)

grep

sed

awk

sort

cut

Episode 3.06

Episode title: **Editors & Productivity**

Objective: 1.5

Linux Text Editors Overview

vi / vim

nano

Choosing the right editor

Text Editors – Exam Reminder

vi / vim – Modal editor

nano – Modeless editor

Know how to:

- Open files
- Save and quit

vi/vim Fundamentals

Modes

Navigation

Saving and quitting safely

nano Fundamentals

Editing

Search

On-screen shortcuts

Text Editors & Productivity (DEMO)

Open files

Edit safely

Save, quit, and verify